The Standard Quantum Limit in Measurement Physics

A Scaffolded Treatise

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Abstract

The standard quantum limit (SQL) is a family of precision benchmarks whose form depends on the measurement model, resource count, probe state, readout mechanism, and noise convention. In continuous position sensing, the conventional SQL follows from optimizing the tradeoff between readout imprecision and quantum backaction for an uncorrelated meter. For phase estimation, the same term commonly describes the $N^{-1/2}$ uncertainty scaling of independent probes. In phase preserving amplification, it denotes the minimum input referred noise required to preserve bosonic commutators. This article develops these cases from calibrated measurement records and spectral densities, then connects linear response theory and quantum noise constraints with Fisher information and the quantum Cramér–Rao bound. The analysis distinguishes imprecision from total sensitivity, tracks single sided and double-sided spectral conventions, and explains how squeezing, correlations, variational readout, speed meters, backaction evasion, entanglement, and quantum nondemolition measurements alter the assumptions behind a chosen benchmark. Applications include gravitational wave interferometry, cavity optomechanics, quantum imaging, force microscopy, spin ensembles, atomic clocks, and microwave amplification.

Contents

1 Scope, vocabulary, and historical calibration

The following assumes familiarity with expectation values, variances, canonical commutators, and elementary Fourier transforms. Density matrices and Fisher information are introduced in later sections before they enter the analysis.

Precision measurement begins with a physical parameter (θ), such as displacement, force, strain, phase, frequency, or magnetic field, whose influence is encoded in a responsive physical system. Depending on the experiment, that response may appear as the motion of a mirror, a shift in a resonator, a change in an optical or microwave mode, or the collective evolution of an atomic or spin ensemble. A transduction mechanism then converts the physical response into an observable classical record, such as a voltage trace or a sequence of photon counts. Finally, a calibrated estimator interprets that record to produce an estimate ($\hat{\theta}$) together with its associated uncertainty.

The term standard quantum limit acquires a meaning once this measurement chain, its available resources, and its noise conventions have been specified. In a conventional continuous position measurement, the SQL arises from the balance between measurement imprecision and quantum backaction. For an ensemble of independent probes, it appears as the statistical scaling in which the estimation uncertainty decreases in proportion to $N^{-1/2}$. In a phase preserving amplifier, the corresponding quantum limit constrains the added noise according to the bosonic commutation relations. These limits may arise within the same laboratory system, although a meaningful numerical comparison requires a consistent definition of the measured quantity, bandwidth, resource count, and spectral convention [1–3].

Table 1: Four uses of quantum limit language that require separate assumptions.

Setting	Bounded quantity	Assumptions that produce the standard benchmark	Common expression
Continuous position meter	Added displacement or force spectrum	Linear response, minimum uncertainty meter, and zero useful imprecision and backaction correlation	$\bar{S}_{xx}^{\mathrm{SQL}} = \hbar\lvert\chi_m\rvert$
Statistical phase estimation	Estimator standard deviation	Independent probes, additive generator, and fixed single probe resource	$\Delta\theta \propto N^{-1/2}$
Phase preserving amplifier	Input referred added noise	Deterministic linear gain for both field quadratures	$n_{\mathrm{add}} \geq 1/2$ at large gain
Correlation aware force sensing	Added force spectrum	Stationary linear sensing with optimized correlations and dissipative response	$\bar{S}_{FF}^{\mathrm{add}} \geq \hbar\lvert \operatorname{Im}\chi_m^{-1}\rvert$

The modern theory of quantum limited measurement grew from Braginskii's analysis of the classical and quantum constraints governing the detection of weak disturbances in a macroscopic oscillator, first published in Russian in 1967 and translated into English in 1968 [1]. Braginsky and Vorontsov subsequently extended this work into a broader theory of measurement limits for macroscopic quantum systems, and Braginsky and Manukin consolidated the associated methods of weak force detection in a monograph [4, 5]. The search for oscillator observables that could be measured repeatedly without disrupting their subsequent evolution led to the theory of quantum nondemolition measurement. Its relevance to gravitational wave detection was articulated by Braginsky, Vorontsov, and Thorne and later examined systematically in a detailed review by Caves and collaborators [2, 6, 7].

Caves clarified the distinct roles of photon counting uncertainty and radiation pressure fluctuations in interferometric measurements, then showed how squeezed optical input could surpass the sensitivity attainable with coherent states. He also established the quantum limit on the added noise of a phase preserving amplifier [8–10]. Subsequent reviews placed these results within broader theories of linear response, cavity optomechanics, precision amplification, and gravitational wave detection, linking the early measurement limits to the concepts and experimental architectures used in modern quantum sensing [3, 11–15]. Related work extended quantum limited estimation to imaging, where spatial information and resolution are themselves treated as measurable quantities subject to statistical and quantum constraints [16].

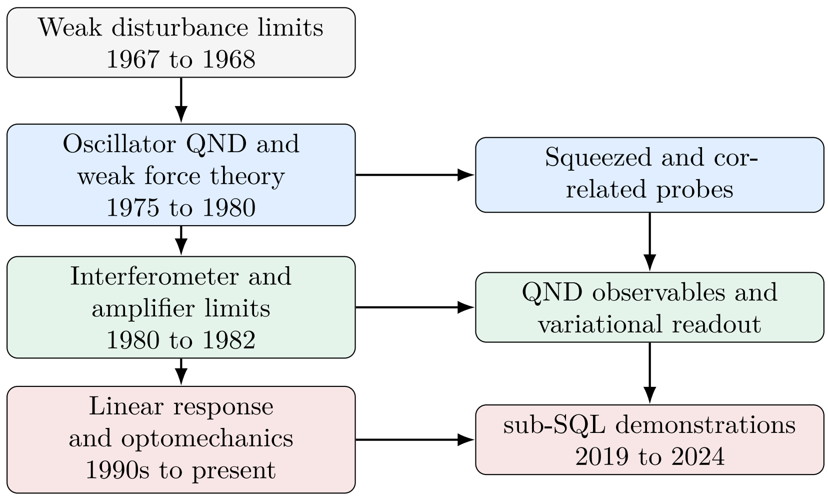

Figure 1: Conceptual lineage of the SQL. The arrows indicate dependence among ideas and carry no claim of exclusive authorship.

Table 2: Core vocabulary used throughout the article.

Term	Meaning
Imprecision	Noise referred to the input that obscures the measured variable in the readout record. Optical phase fluctuations and amplifier output noise are common sources.
Backaction	Fluctuating disturbance exerted by the meter on the measured system. Radiation pressure force and charge force noise are common examples.
Susceptibility	Complex linear response $\chi_m(\Omega)$ that maps applied force to displacement. Its magnitude sets signal gain, and its phase sets the correlation needed for cancellation.
Spectral density	Frequency resolved covariance of stationary fluctuations. Its numerical value depends on sidedness, symmetrization, and Fourier normalization.
Coherent state	Minimum uncertainty optical state whose two quadratures have equal vacuum variance. It supplies the common laser reference.
Squeezed state	State with reduced variance in one field quadrature and increased variance in its conjugate quadrature.
Quantum nondemolition measurement	Repeated measurement of an observable whose future values remain compatible with earlier measurements under the relevant dynamics and coupling.
Backaction evasion	Coupling or readout design that directs disturbance away from the signal observable or cancels its contribution in the estimator.

Term	Meaning
Quantum Fisher information	Largest local Fisher information available from a parameter dependent quantum state after optimization over quantum measurements.

Practical rule of thumb

Write the measured variable, resource, probe state, readout, bandwidth, and spectral convention beside every SQL formula. An unlabeled SQL curve carries incomplete physical information.

2 Measurement records, response functions, and spectra

The following assumes familiarity with complex Fourier amplitudes and the linear response of a driven harmonic oscillator. The symmetrized spectral density is introduced below, so no previous experience with quantum noise spectra is required.

2.1 From a physical disturbance to an estimator

A linear sensor can be understood as a chain of transduction steps. An external force perturbs a mechanical coordinate, whose response is acquired by an optical or electrical probe. The detector converts the probe response into a classical signal, and a filter processes that signal to estimate the applied force or another parameter of interest. Every stage contributes gain, bandwidth, loss, and noise. Expressing these contributions in terms of a common input variable allows their fluctuations and correlations to be combined within a single covariance model.

Let $x(t)$ denote the displacement of a mechanical mode and $F_{\mathrm{sig}}(t)$ the force to be estimated. The Fourier domain response is

$$x(\Omega) = \chi_m(\Omega)\left[F_{\mathrm{sig}}(\Omega) + F_{\mathrm{ba}}(\Omega) + F_{\mathrm{th}}(\Omega)\right], \tag{1}$$

where F_{ba} is meter backaction and F_{th} represents environmental force noise. For an oscillator with effective mass m, resonance Ω_m, and energy damping rate Γ_m,

$$\chi_m(\Omega) = \frac{1}{m\left(\Omega_m^2 - \Omega^2 - i\Gamma_m\Omega\right)}. \tag{2}$$

The free mass approximation follows from $\Omega_m = 0$ and negligible damping over the signal band, which gives $\chi_m = -1/(m\Omega^2)$.

After calibration to displacement, the measured record is

$$y(\Omega) = x(\Omega) + x_{\mathrm{imp}}(\Omega), \tag{3}$$

where x_{imp} is input referred imprecision. Equations (1) and (3) form the minimal model behind the continuous measurement SQL.

2.2 Symmetrized double-sided spectra

For stationary Hermitian fluctuations $\delta\hat{A}$ and $\delta\hat{B}$, this article uses

$$\bar{S}_{AB}(\Omega) = \frac{1}{2}\int_{-\infty}^{\infty} \mathrm{d}t\, e^{i\Omega t} \left\langle \left\{ \delta\hat{A}(t), \delta\hat{B}(0) \right\} \right\rangle . \tag{4}$$

The bar denotes a symmetrized double-sided spectrum per unit angular frequency. For an auto-spectrum, or for an even classical symmetrized spectrum, the corresponding one-sided spectrum defined for $\Omega > 0$ is

$$S_{AB}^{(1)}(\Omega) = 2\bar{S}_{AB}(\Omega). \tag{5}$$

An amplitude spectral density is the square root of a power spectral density. Displacement amplitude spectra carry units of $\mathrm{m}/\sqrt{\mathrm{Hz}}$, force amplitude spectra carry $\mathrm{N}/\sqrt{\mathrm{Hz}}$, and strain amplitude spectra carry $1/\sqrt{\mathrm{Hz}}$ after conversion to spectra per hertz.

Convention checkpoint

Every factor of two in an SQL spectrum depends on sidedness and symmetrization. Equations in this article use $\bar{S}$ for symmetrized double-sided spectra. A one sided result follows from Eq. (5).

At each angular frequency, the meter noise is characterized by the spectral density matrix

$$\bar{\mathbf{S}}_m(\Omega) = \begin{pmatrix} \bar{S}_{xx}^{\mathrm{imp}} & \bar{S}_{xF} \\ \bar{S}_{xF}^{*} & \bar{S}_{FF}^{\mathrm{ba}} \end{pmatrix} . \tag{6}$$

Referring the measurement record to an equivalent oscillator displacement yields the added displacement noise spectrum

$$\bar{S}_{xx}^{\mathrm{add}}(\Omega) = \bar{S}_{xx}^{\mathrm{imp}} + |\chi_m(\Omega)|^2 \bar{S}_{FF}^{\mathrm{ba}} + 2\,\mathrm{Re}\Big[\chi_m^{*}(\Omega)\bar{S}_{xF}\Big] . \tag{7}$$

After all noise contributions have been expressed in displacement units, the measured spectrum is

$$\bar{S}_{xx}^{\mathrm{meas}} = \bar{S}_{xx}^{\mathrm{intrinsic}} + \bar{S}_{xx}^{\mathrm{add}} + \bar{S}_{xx}^{\mathrm{tech}}, \tag{8}$$

where $\bar{S}_{xx}^{\mathrm{intrinsic}}$ contains the equilibrium mechanical motion, including its thermal and zero-point contributions.

The cross spectrum $\bar{S}_{xF}$ between measurement imprecision and backaction controls the correlation contribution to the added noise. A phase quadrature readout of a coherent probe often provides little exploitable correlation. Variational homodyne detection can instead select a readout quadrature that exploits correlations associated with ponderomotive squeezing, allowing the correlation term in Eq. (7) to become negative over a selected frequency range and thereby reduce the added displacement noise.

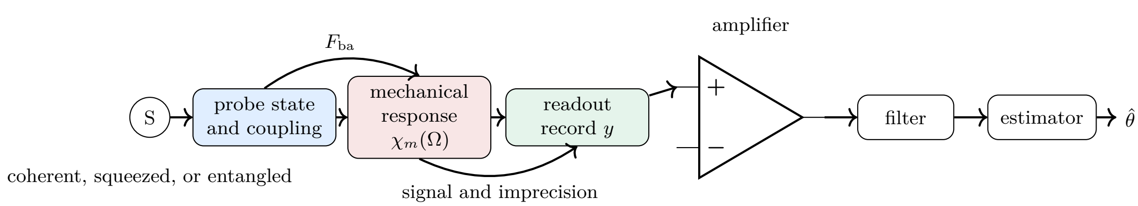

Figure 2: A calibrated quantum measurement chain. SQL performance depends on the full chain and on correlations retained by the estimator.

Practical rule of thumb

Refer every noise source to one variable before addition. Preserve the cross spectral terms until the chosen estimator and readout quadrature have been fixed.

3 Derivation of the continuous position and force SQL

Review Eqs. (2), (6), and (7). The derivation uses a minimum uncertainty linear meter with negligible reverse gain and states every spectral convention explicitly.

3.1 Quantum noise constraint

A general linear detector has output operator $\hat{I}$, backaction operator $\hat{F}$, forward susceptibility χ_{IF}, and reverse susceptibility χ_{FI}. Define

$$\widetilde{\chi}_{IF}(\Omega) = \chi_{IF}(\Omega) - \chi^*_{FI}(\Omega). \tag{9}$$

The finite frequency quantum noise theorem can be written

$$\bar{S}_{II}\bar{S}_{FF} - |\bar{S}_{IF}|^2 \geq \left|\frac{\hbar\widetilde{\chi}_{IF}}{2}\right|^2 \left[1 + \Delta\left(\frac{\bar{S}_{IF}}{\hbar\widetilde{\chi}_{IF}/2}\right)\right], \tag{10}$$

where

$$\Delta[z] = \frac{|1+z^2| - (1+|z|^2)}{2}. \tag{11}$$

For the uncorrelated detector model used in the following SQL derivation, $\bar{S}_{IF} = 0$ and hence $\Delta[0] = 0$. The complete finite-frequency inequality remains necessary for detectors with general complex correlations, reverse gain, or nontrivial dynamical response [3].

For a canonical meter with negligible reverse gain, input referral by the forward gain and conditions that remove the correction reduce Eq. (10) to

$$\bar{S}^{\text{imp}}_{xx}\bar{S}^{\text{ba}}_{FF} - |\bar{S}_{xF}|^2 \geq \frac{\hbar^2}{4}. \tag{12}$$

The interaction Hamiltonian and output response are

$$\hat{H}_{\text{int}} = -\hat{x}\hat{F}, \qquad \hat{I}(\Omega) = \hat{I}_0(\Omega) + \chi_{IF}(\Omega)\hat{x}(\Omega). \tag{13}$$

Equation (12) supplies the transparent algebra for the standard uncorrelated SQL. The full theorem is required for a general detector or a realizability proof [17, 18].

3.2 Optimization for an uncorrelated meter

Set $\bar{S}_{xF} = 0$ and parameterize measurement strength by a positive dimensionless variable λ,

$$\bar{S}^{\text{imp}}_{xx} = \frac{A}{\lambda}, \qquad \bar{S}^{\text{ba}}_{FF} = B\lambda. \tag{14}$$

Stronger coupling carries more information and lowers imprecision, and the same coupling raises the fluctuating force. The added displacement spectrum is

$$\bar{S}_{xx}^{\mathrm{add}} = \frac{A}{\lambda} + |\chi_m|^2 B\lambda. \tag{15}$$

Differentiation with respect to λ gives

$$\lambda_{\mathrm{opt}} = \sqrt{\frac{A}{|\chi_m|^2 B}}, \tag{16}$$

and

$$\bar{S}_{xx,\min}^{\mathrm{add}} = 2|\chi_m|\sqrt{AB}. \tag{17}$$

For a meter that saturates Eq. (12), $AB = \hbar^2/4$, which yields

$$\boxed{\bar{S}_{xx}^{\mathrm{SQL}}(\Omega) = \hbar|\chi_m(\Omega)|}. \tag{18}$$

The force referred form follows from division by $|\chi_m|^2$,

$$\boxed{\bar{S}_{FF}^{\mathrm{SQL}}(\Omega) = \frac{\hbar}{|\chi_m(\Omega)|}}. \tag{19}$$

At the optimum of this uncorrelated model, the imprecision and backaction contributions are equal; other architectures can reach their optima with unequal contributions.

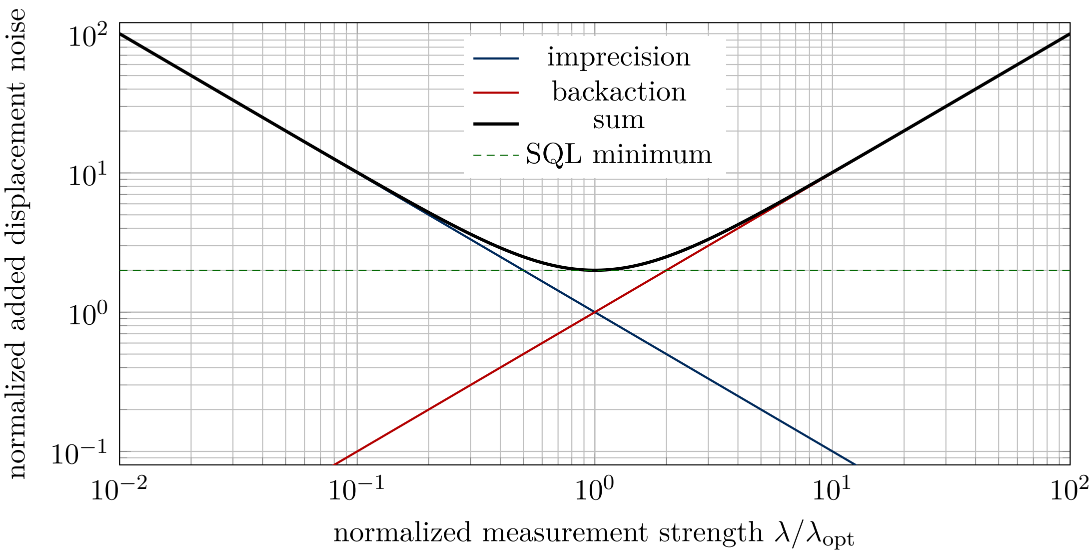

Figure 3: Normalized imprecision and backaction tradeoff for an uncorrelated linear meter.

3.3 Free mass and resonant oscillator

For a free mass, $|\chi_m| = 1/(m\Omega^2)$, so

$$\bar{S}_{xx}^{\mathrm{SQL}} = \frac{\hbar}{m\Omega^2}, \qquad \bar{S}_{FF}^{\mathrm{SQL}} = \hbar m\Omega^2. \tag{20}$$

For a lightly damped oscillator at resonance,

$$|\chi_m(\Omega_m)| = \frac{1}{m\Gamma_m\Omega_m}, \qquad \bar{S}_{FF}^{\mathrm{SQL}}(\Omega_m) = \hbar m\Gamma_m\Omega_m. \tag{21}$$

One sided spectra are twice these values. The square root amplitude spectra therefore acquire a factor $\sqrt{2}$.

The equilibrium one sided thermal force spectrum of a viscously damped oscillator can be written

$$S_{FF}^{\mathrm{th},(1)}(\Omega) = 2m\Gamma_m \hbar\Omega \coth\left(\frac{\hbar\Omega}{2k_B T}\right). \tag{22}$$

Its classical limit is $4m\Gamma_m k_B T$. Quantum measurement noise becomes experimentally visible after thermal, laser, electronic, and calibration contributions approach the relevant SQL spectrum.

3.4 Frequency dependent tuning landscape

The oscillator susceptibility changes rapidly near resonance, so a fixed probe strength reaches the optimum over a limited band. Define

$$\widetilde{\chi}(r) = \frac{1}{\sqrt{(1-r^2)^2 + (\gamma r)^2}}, \qquad r = \frac{\Omega}{\Omega_m}, \tag{23}$$

and normalized added noise

$$N(r,\lambda) = \lambda^{-1} + \lambda|\widetilde{\chi}(r)|^2. \tag{24}$$

With $u = \lambda|\widetilde{\chi}|$, the SQL ratio is

$$\frac{N}{N_{\mathrm{SQL}}} = \frac{1}{2}\left(u + u^{-1}\right) \geq 1. \tag{25}$$

Equality gives $\lambda_{\mathrm{opt}} = 1/|\widetilde{\chi}|$. Figure 4 evaluates this expression directly with PGFPlots and requires no external data file.

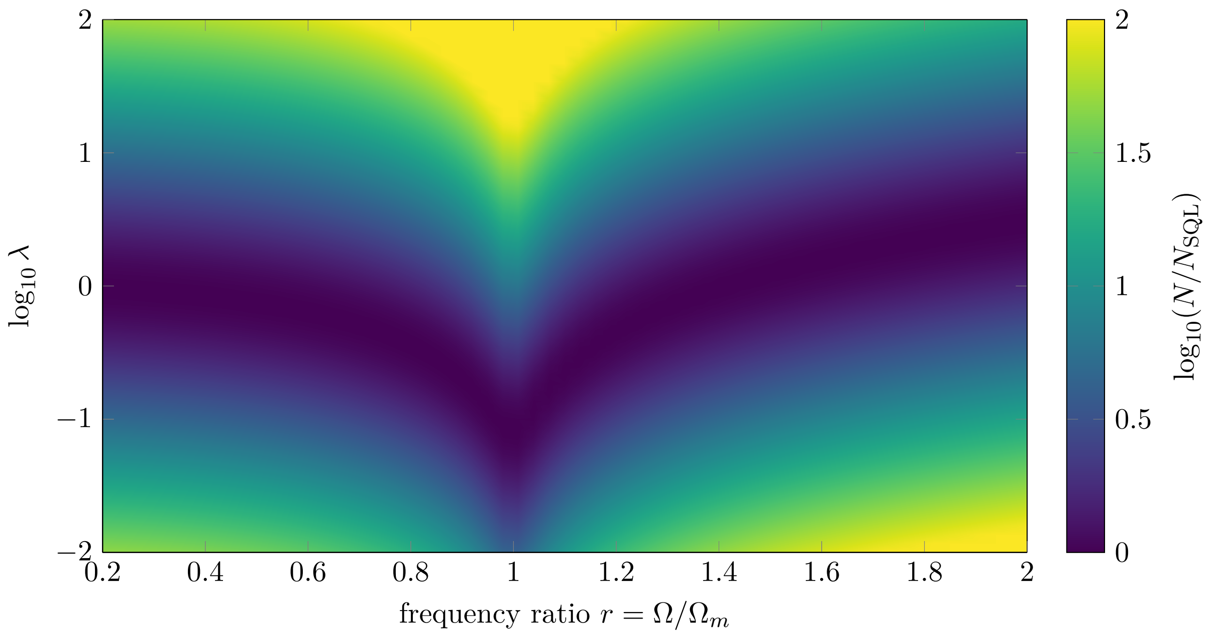

Figure 4: Excess added noise above the uncorrelated SQL for damping ratio $\gamma = 0.08$. The valley follows $\lambda_{\mathrm{opt}} = 1/|\widetilde{\chi}|$, and the plotted surface is generated analytically from the provided mathematical formulas to ensure computational reproducibility.

3.5 SQL and the correlation aware lower bound

Useful $\bar{S}_{xF}$ can lower Eq. (7) below Eq. (18). Stationary linear force sensing remains constrained by dissipation. In the same double-sided convention, common forms of the ultimate added noise bounds are

$$\bar{S}_{xx}^{\mathrm{add}} \geq \hbar |\operatorname{Im} \chi_m|, \qquad \bar{S}_{FF}^{\mathrm{add}} \geq \hbar |\operatorname{Im} \chi_m^{-1}|. \tag{26}$$

For viscous damping, the force bound reduces to the dissipative quantum limit (DQL),

$$\bar{S}_{FF}^{\mathrm{DQL}}(\Omega) = \hbar m \Gamma_m |\Omega|. \tag{27}$$

At mechanical resonance χ_m is nearly imaginary, so the uncorrelated displacement SQL and the dissipative displacement bound coincide. Away from resonance, their separation quantifies the improvement available from correlations. An ideal lossless free mass has a real susceptibility, and Eq. (26) then signals the importance of finite observation time, finite probe energy, optical loss, and other resource constraints that lie outside the stationary dissipative bound [19–21].

Practical rule of thumb

Balance imprecision and backaction to locate the uncorrelated SQL. Use the full covariance matrix and the dissipative response to evaluate any claimed advantage from correlations.

4 Optical interferometers and cavity optomechanics

This section assumes familiarity with optical phase, coherent state photon statistics, and the susceptibility of a harmonic oscillator. The spectral relations needed for the discussion were developed in Section 2.

4.1 Coherent light in a Michelson position meter

A differential arm displacement x changes the round trip phase by

$$\phi = 2k_0 x = \frac{4\pi x}{\lambda_0}. \tag{28}$$

For N detected photons in a coherent state, phase imprecision scales as

$$\Delta\phi_{\mathrm{imp}} \sim N^{-1/2}, \qquad \Delta x_{\mathrm{imp}} \sim \frac{\lambda_0}{4\pi\sqrt{N}}. \tag{29}$$

With optical power P, integration time τ, and carrier frequency ω_0, the detected photon number scales as $N \sim P\tau/(\hbar\omega_0)$. The displacement imprecision spectrum therefore scales as P^{-1}.

Coherent state amplitude quadrature fluctuations produce photon flux fluctuations around a bright carrier. Their fluctuating momentum transfer produces a radiation pressure force spectrum proportional to P. The phase quadrature produces readout imprecision. The equivalent displacement spectrum from force noise is multiplied by $|\chi_m|^2$, so a compact model is

$$\bar{S}_{xx}^{\mathrm{add}}(\Omega; P) = \frac{\alpha(\Omega)}{P} + \beta(\Omega) P |\chi_m(\Omega)|^2. \tag{30}$$

The optical commutator fixes $\alpha\beta$ for an ideal coherent probe. Minimization over P reproduces Eq. (18). Strain calibration in a simple differential arm model uses $x = hL$, which gives $S_{hh} = S_{xx}/L^2$ after the effective differential mechanical mode has been defined.

The full interferometer input and output relations include arm cavities, signal recycling, photodetection loss, homodyne angle, and optomechanical coupling. Their frequency dependence rotates the output noise ellipse. Filter cavities can rotate an injected squeezed ellipse to track that response, which reduces high frequency phase noise and low frequency radiation pressure noise in one broadband configuration [22–24].

4.2 Cavity optomechanical implementation

For a single optical and mechanical mode, the dispersive Hamiltonian is

$$\hat{H} = \hbar\omega_c \hat{a}^\dagger \hat{a} + \hbar\Omega_m \hat{b}^\dagger \hat{b} - \hbar g_0 \hat{a}^\dagger \hat{a} \left(\hat{b} + \hat{b}^\dagger\right). \tag{31}$$

The single photon coupling g_0 gives the cavity frequency shift produced by zero point motion. Driving the cavity to mean photon number n_c produces the linearized coupling $G = g_0\sqrt{n_c}$. For a resonant drive in the unresolved sideband measurement limit, a representative measurement rate is

$$\Gamma_{\text{meas}} \simeq \frac{4\eta G^2}{\kappa}, \tag{32}$$

where η is total detection efficiency and κ is the cavity energy decay rate. Larger n_c increases information rate and radiation pressure shot noise together. The cooperativity

$$C = \frac{4G^2}{\kappa\Gamma_m} \tag{33}$$

compares optomechanical measurement dynamics with mechanical damping. Quantum cooperativity also accounts for thermal occupation and indicates whether quantum backaction can compete with thermalization [14, 25].

The oscillator zero point displacement is

$$x_{\text{zpf}} = \sqrt{\frac{\hbar}{2m\Omega_m}}. \tag{34}$$

For $k_B T \gg \hbar\Omega_m$, the thermal occupation is $n_{\text{th}} \simeq k_B T/(\hbar\Omega_m)$. The product $\Gamma_m n_{\text{th}}$ estimates a thermal heating rate. At fixed quality factor $Q = \Omega_m/\Gamma_m$,

$$\Gamma_m n_{\text{th}} \simeq \frac{k_B T}{\hbar Q}. \tag{35}$$

Frequency alone therefore gives no reduction of this rate at fixed Q. Material loss, temperature, surface contamination, optical absorption, and mode shape determine the realized thermal environment.

Experiments have observed radiation pressure backaction, measured forces close to the SQL, and used optomechanical correlations to obtain total off resonant force and displacement sensitivity below the uncorrelated SQL [19, 26, 27]. The last statement is stronger than a report of imprecision below an SQL scale. It includes the calibrated intrinsic mechanical motion, measurement imprecision, backaction, and their correlations in the measured band.

Practical rule of thumb

Increasing optical power improves phase readout until radiation pressure, absorption, heating, or dynamical backaction erases the gain. Track detected efficiency, intracavity power, and mechanical thermalization as separate control variables.

5 From continuous records to Fisher information

This section assumes a basic understanding of density matrices ρ, including their role as positive, unit trace operators that describe pure and mixed quantum states. Familiarity with conditional probability and variance will also be useful. The estimators and Fisher information quantities needed later are introduced as they arise.

5.1 Classical estimators and local information

A measurement with outcomes k and parameter dependent probabilities $p(k|\theta)$ has classical Fisher information

$$F_C(\theta) = \sum_k p(k|\theta) \left[\frac{\partial}{\partial \theta} \ln p(k|\theta) \right]^2 . \tag{36}$$

For ν independent repetitions, a locally unbiased estimator obeys the Cramér–Rao inequality

$$\mathrm{Var}(\hat{\theta}) \geq \frac{1}{\nu F_C(\theta)}. \tag{37}$$

The bound describes local precision after the likelihood, measurement, and regularity conditions have been fixed. Bias, nuisance parameters, finite samples, broad priors, and periodic parameters require corresponding extensions.

A quantum state ρ_θ and a positive operator valued measure $\{E_k\}$ give

$$p(k|\theta) = \mathrm{Tr}(\rho_\theta E_k), \qquad E_k \succeq 0, \qquad \sum_k E_k = \mathbb{I}. \tag{38}$$

Optimization over all measurements yields quantum Fisher information. The symmetric logarithmic derivative L_θ satisfies

$$\partial_\theta \rho_\theta = \frac{1}{2} \left(L_\theta \rho_\theta + \rho_\theta L_\theta \right), \tag{39}$$

and

$$F_Q(\theta) = \mathrm{Tr}(\rho_\theta L_\theta^2), \qquad F_C(\theta) \leq F_Q(\theta). \tag{40}$$

The quantum Cramér–Rao bound is

$$\mathrm{Var}(\hat{\theta}) \geq \frac{1}{\nu F_Q(\theta)}. \tag{41}$$

Attainability depends on the state family, available measurement, local operating point, and multiparameter compatibility [28–30].

For unitary encoding generated by $\hat{G}$ and a mixed state $\rho = \sum_j p_j |j\rangle\langle j|$,

$$F_Q[\rho, \hat{G}] = 2 \sum_{j,k: p_j + p_k > 0} \frac{(p_j - p_k)^2}{p_j + p_k} |\langle j|\hat{G}|k\rangle|^2. \tag{42}$$

For a pure state $|\psi\rangle$, this expression reduces to

$$F_Q = 4(\Delta\hat{G})^2_\psi. \tag{43}$$

Quantum Fisher information quantifies available information. Probe preparation, entanglement, squeezing, interaction engineering, and measurement design supply the physical resources that change it.

5.2 Independent probes and the statistical SQL

Let the parameter be encoded by

$$\hat{U}_\theta = e^{-i\theta\hat{G}}, \qquad \hat{G} = \sum_{j=1}^{N} \hat{g}_j. \tag{44}$$

For N independent probes in a product state, local generator variances add,

$$(\Delta\hat{G})^2 = N(\Delta\hat{g})^2, \qquad F_Q \propto N, \qquad \Delta\theta \propto N^{-1/2}. \tag{45}$$

If the jth generator has spectral range $\Delta g_j = g_{j,\max} - g_{j,\min}$, every separable state obeys

$$F_Q[\rho_{\text{sep}}, \hat{G}] \leq \sum_{j=1}^{N} (\Delta g_j)^2. \tag{46}$$

Identical qubits with $\hat{g}_j = \sigma_z^{(j)}/2$ satisfy $F_Q \leq N$, so

$$\mathrm{Var}(\hat{\theta}) \geq \frac{1}{\nu N}. \tag{47}$$

This inequality gives a precise restricted state QCRB and supports the phrase statistical SQL [31, 32].

An ideal GHZ state places amplitude at the extremal eigenvalues of the total generator and reaches

$$F_Q = N^2, \qquad \mathrm{Var}(\hat{\theta}) \geq \frac{1}{\nu N^2}. \tag{48}$$

This Heisenberg scaling assumes a consistent resource count and coherent local encoding. Sequential coherent reuse can reproduce the same scaling under a resource definition that counts every pass. Independent loss and Markovian dephasing commonly return the asymptotic scaling toward $N^{-1/2}$, leaving a useful constant factor advantage over separable probes [33–35].

5.3 The bridge from spectra to estimators

The preceding continuous measurement theory describes a noisy record indexed by time or Fourier frequency. Fisher information describes the distinguishability of records or states indexed by a parameter. A stationary Gaussian record connects the two descriptions directly.

Consider

$$y(\Omega) = \theta s(\Omega) + n(\Omega), \tag{49}$$

where s is a known finite duration signal template and n is zero mean Gaussian noise with double-sided spectrum $\bar{S}_{nn}$. The matched filter estimator is

$$\hat{\theta} = \frac{\displaystyle\int_{-\infty}^{\infty} \frac{\mathrm{d}\Omega}{2\pi} \frac{s^*(\Omega) y(\Omega)}{\bar{S}_{nn}(\Omega)}}{\displaystyle\int_{-\infty}^{\infty} \frac{\mathrm{d}\Omega}{2\pi} \frac{|s(\Omega)|^2}{\bar{S}_{nn}(\Omega)}}. \tag{50}$$

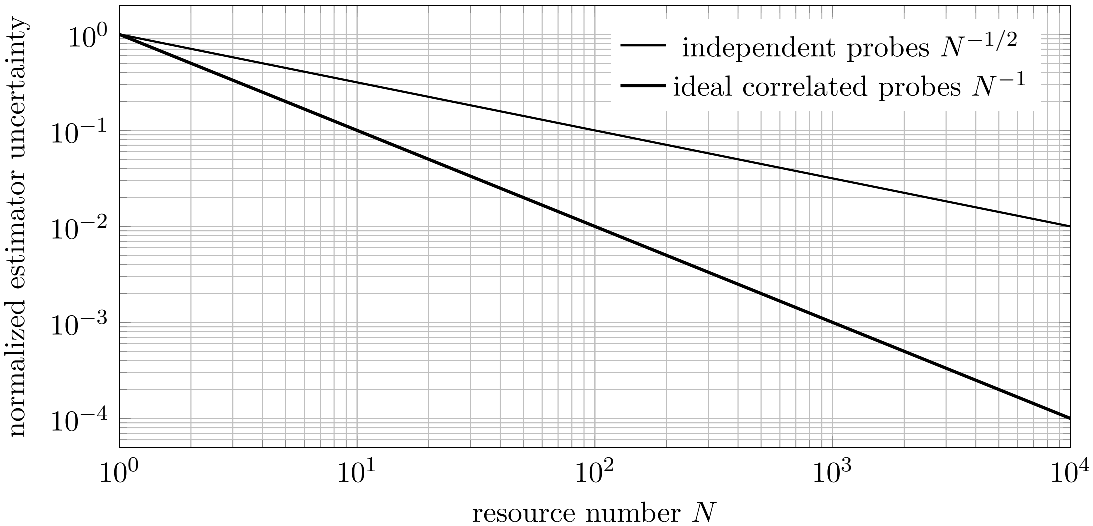

Figure 5: Ideal resource scaling. Loss, decoherence, and preparation overhead determine the realized advantage.

Its variance is

$$\mathrm{Var}(\hat{\theta}) = \left[\int_{-\infty}^{\infty} \frac{\mathrm{d}\Omega}{2\pi} \frac{|s(\Omega)|^2}{\bar{S}_{nn}(\Omega)}\right]^{-1}, \tag{51}$$

which is the inverse Fisher information for this Gaussian model. Force estimation uses $s(\Omega) = \chi_m(\Omega)F_0(\Omega)$ after displacement calibration. Strain estimation uses the interferometer response to the chosen waveform.

Probe strength changes $\bar{S}_{nn}$ by changing imprecision, backaction, and their cross spectrum. Minimizing a narrowband noise spectrum therefore maximizes the Fisher information for a signal in that band. Broadband optimization weights every frequency by $|s|^2/\bar{S}_{nn}$, so the smallest pointwise spectrum and the smallest waveform estimation error can select different instrument settings. Functional quantum bounds extend this logic to time dependent forces and show how quantum noise cancellation and smoothing can approach the waveform QCRB [36]. The statistical and continuous SQLs meet at this estimator layer.

Practical rule of thumb

Use $F_Q \propto N$ as the precise separable probe benchmark. Use matched filter Fisher information for a continuous waveform. Keep resource counting, bandwidth, and estimator bias fixed during every comparison.

6 Physical routes below the standard benchmark

Review the meter covariance matrix in Eq. (6), the added noise expression in Eq. (7), and the distinction between the SQL and Eq. (26).

6.1 Squeezed quadratures

Define optical quadratures by

$$\hat{X}_1 = \frac{\hat{a} + \hat{a}^\dagger}{2}, \qquad \hat{X}_2 = \frac{\hat{a} - \hat{a}^\dagger}{2i}, \qquad [\hat{X}_1, \hat{X}_2] = \frac{i}{2}. \tag{52}$$

Vacuum and coherent states have variance 1/4 in each quadrature. A squeezed Gaussian state has covariance

$$\mathbf{V}(r,\phi) = \frac{1}{4}\mathbf{R}(\phi)\begin{pmatrix} e^{-2r} & 0 \\ 0 & e^{2r} \end{pmatrix}\mathbf{R}^{\mathsf{T}}(\phi), \tag{53}$$

where r is squeeze strength and ϕ is squeeze angle. A fixed phase squeezed input lowers high frequency readout imprecision and raises the conjugate amplitude fluctuations that drive low frequency radiation pressure. Frequency dependent rotation $\phi(\Omega)$ aligns the reduced variance with the frequency dependent signal quadrature [9, 22, 23].

Loss replaces a fraction of the state with vacuum,

$$\mathbf{V}_{\mathrm{obs}} = \eta\mathbf{V}(r,\phi) + (1-\eta)\frac{\mathbf{I}}{4}. \tag{54}$$

Large antisqueezing also converts small phase jitter into observed noise. Propagation loss, mode mismatch, photodetection efficiency, filter cavity loss, and squeeze angle control therefore set the realized improvement.

6.2 Correlation engineering and variational readout

Choose the phase of $\bar{S}_{xF}$ so that

$$\mathrm{Re}\Big[\chi_m^* \bar{S}_{xF}\Big] < 0. \tag{55}$$

The cross term in Eq. (7) then removes part of the backaction contribution from the record. Completing the square gives

$$\bar{S}_{xx}^{\mathrm{add}} = \left(\bar{S}_{xx}^{\mathrm{imp}} - \frac{|\bar{S}_{xF}|^2}{\bar{S}_{FF}^{\mathrm{ba}}}\right) + \bar{S}_{FF}^{\mathrm{ba}}\left|\chi_m + \frac{\bar{S}_{xF}}{\bar{S}_{FF}^{\mathrm{ba}}}\right|^2. \tag{56}$$

Within the reduced detector model, Eq. (12) gives

$$\bar{S}_{xx}^{\mathrm{add}} \geq \frac{\hbar^2}{4\bar{S}_{FF}^{\mathrm{ba}}} + \bar{S}_{FF}^{\mathrm{ba}}\left|\chi_m + \frac{\bar{S}_{xF}}{\bar{S}_{FF}^{\mathrm{ba}}}\right|^2. \tag{57}$$

Useful anticorrelation lowers the second term. The simplified determinant constraint does not determine the physically realizable broadband optimum. Detector gain, reverse gain, causality, dynamical backaction, optical loss, available coupling, and realizable homodyne angles constrain $\bar{S}_{xF}$.

Variational readout implements this idea by selecting a frequency dependent homodyne quadrature. Ponderomotive interaction creates correlations between optical amplitude and phase, and the readout selects the combination that retains signal with reduced backaction contamination. Broadband displacement experiments have used these correlations to move below the ordinary uncorrelated sensitivity curve [19, 37].

Coherent quantum noise cancellation introduces an auxiliary response with an effective sign opposite to the sensor response. Proper matching cancels backaction at the level of coherent dynamics, which is closely related to negative mass reference frames and quantum mechanics free subsystems [38].

6.3 Quantum nondemolition observables and backaction evasion

An ideal QND observable $\hat{A}$ satisfies

$$[\hat{A}(t_i), \hat{A}(t_j)] = 0 \tag{58}$$

at the sampling times, and the measurement interaction preserves that observable. A sufficient ideal coupling condition is

$$[\hat{A}, \hat{H}_{\text{int}}] = 0. \tag{59}$$

The disturbance enters a conjugate variable whose uncertainty is absent from the later target measurement [6, 7, 12].

For a harmonic oscillator, define slowly varying rotating quadratures $\hat{X}$ and $\hat{P}$ with $[\hat{X}, \hat{P}] = i$. A balanced two tone interaction can couple the meter to one rotating quadrature. Backaction enters the conjugate quadrature, so repeated measurements of the selected quadrature evade the ordinary position SQL over the rotating wave and bandwidth regime. Counterrotating terms, damping, imperfect sideband resolution, and pump imbalance leak backaction into the measured quadrature. Experiments have implemented QND and backaction evading protocols in microwave and optical optomechanical systems [39–41].

For a free mass, its later position depends on the momentum it carried earlier. Any momentum disturbance introduced by the first measurement therefore appears in subsequent position readouts as backaction noise. A speed meter senses a velocity like observable, and variational readout estimates a combination of output quadratures whose backaction contribution is reduced. Contractive state analyses and realizable measurement models also show that the free mass SQL depends on state preparation and measurement protocol [42, 43]. Equation (20) should therefore be identified as the SQL of the stated uncorrelated continuous position meter.

6.4 Phase preserving amplification

A deterministic phase preserving bosonic amplifier with power gain G has the canonical map

$$\hat{a}_{\text{out}} = \sqrt{G}\,\hat{a}_{\text{in}} + \sqrt{G-1}\,\hat{b}_{\text{in}}^{\dagger}. \tag{60}$$

The auxiliary mode preserves $[\hat{a}_{\text{out}}, \hat{a}_{\text{out}}^{\dagger}] = 1$. With the auxiliary input in vacuum, the minimum input referred added noise is

$$n_{\text{add}} \geq \frac{1}{2}\left(1 - \frac{1}{G}\right), \tag{61}$$

which approaches one half quantum at large gain. Phase sensitive amplification can add zero noise to one selected quadrature as it deamplifies the conjugate quadrature. It gives up simultaneous phase preserving gain for both quadratures [10, 44].

Table 3: Representative strategies for surpassing a standard benchmark.

Method	Change to the measurement model	Representative platform
Phase squeezing	Reduces detected phase quadrature variance	Optical phase sensors and interferometers
Frequency dependent squeezing	Rotates the noise ellipse across Fourier frequency	Gravitational wave detectors
Variational readout	Selects a correlated output quadrature that cancels backaction in the estimator	Interferometers and optomechanics
Backaction evasion	Couples the meter to one rotating mechanical quadrature	Microwave and optical optomechanics
Speed meter	Measures a velocity like observable with repeated interaction geometry	Gravitational wave detector concepts

Method	Change to the measurement model	Representative platform
Coherent noise cancellation	Matches an auxiliary response of opposite sign to the sensor response	Optomechanical force sensors
Spin squeezing	Reduces collective spin variance at fixed mean spin signal	Atomic clocks and magnetometers
Entangled probes	Increases total generator variance above separable scaling	Optical and atomic phase estimation
QND measurement	Selects an observable compatible with its future values	Oscillators, photons, spins, and qubits
Phase sensitive amplification	Amplifies one chosen field quadrature	Microwave qubit readout and radio frequency sensing

Practical rule of thumb

Identify the assumption changed by each sub SQL technique. Squeezing changes the probe covariance, variational readout changes the measured quadrature, QND changes the observable, and entanglement changes generator variance.

7 Applications and evidence standards

Review the difference among imprecision, added noise, and total measured noise. A claim about one level gives limited information about the others.

7.1 Gravitational wave interferometry

Laser interferometers infer strain from differential arm motion over kilometer baselines. Optical phase noise dominates much of the high frequency quantum noise, and radiation pressure motion gains importance at lower frequencies. Squeezed vacuum injection first improved detector sensitivity beyond the coherent shot noise level, Advanced LIGO later operated with squeezed vacuum, and frequency dependent squeezing then provided broadband quantum noise reduction [24, 45–47].

The LIGO Livingston analysis reported the quantum noise contribution below the free mass SQL from 35 to 75 Hz with a maximum improvement of 3 dB near 50 Hz after installation of the A+ frequency dependent squeezing system [48, 49]. Classical noise exceeded quantum noise in that low frequency band, so the reported quantum spectrum was inferred by calibrated model subtraction and dedicated squeeze angle measurements. The distinction preserves the importance of the result and states its experimental scope accurately.

7.2 Optomechanical and atomic force sensing

Murch and collaborators measured quantum radiation pressure backaction on the collective motion of an ultracold atomic gas [26]. Schreppler and collaborators observed the imprecision and backaction tradeoff in force sensing and reached a sensitivity consistent with the expected noise model at four times the absolute SQL after residual thermal disturbance and detection efficiency were included [27]. Mason and collaborators used strong optomechanical correlations to obtain off-resonant total force and displacement sensitivity 1.5 dB below the SQL after including thermal noise, measurement imprecision, and backaction in the calibrated noise budget [19].

Atomic force microscopy introduces cantilever thermal noise, optical readout imprecision, laser backaction, tip sample interaction noise, and bandwidth dependent calibration. Early analyses derived quantum and thermal sensitivity constraints for common AFM readouts [50, 51]. Pooser and collaborators used truncated nonlinear interferometry to measure cantilever beam displacement with up to 3 dB quantum noise reduction below the coherent optical reference and reached $1.7\,\mathrm{fm}/\sqrt{\mathrm{Hz}}$ readout sensitivity [52]. This result demonstrates quantum enhanced optical readout. A total mechanical force SQL comparison additionally requires backaction, cantilever thermal noise, and transduction calibration in the same band.

Magnetic resonance force microscopy reached single electron spin detection by coupling spin dynamics to a sensitive cantilever [53]. That milestone concerns signal detection and force sensitivity. Its relation to an SQL requires the complete quantum and thermal noise budget for the chosen force estimator.

7.3 Spin ensembles and clocks

For N uncorrelated two level atoms polarized along one collective spin direction, transverse projection noise scales as $\sqrt{N}$ and phase uncertainty scales as $N^{-1/2}$. The Wineland metrological squeezing parameter is

$$\xi_R^2 = \frac{N(\Delta J_\perp)^2}{|\langle \mathbf{J} \rangle|^2}. \tag{62}$$

Values $\xi_R^2 < 1$ indicate phase sensitivity below the coherent spin state SQL after contrast loss is included. The parameter connects reduced collective variance with a usable signal slope and remains central to atomic clocks, magnetometers, and atom interferometers [30, 54, 55].

7.4 Spatial parameter estimation and quantum imaging

In quantum imaging, the unknown quantity may be a source position, source separation, object displacement, spatial phase, or another parameter encoded in the optical field. Diffraction determines which spatial modes reach the detector, and estimation theory determines how accurately the parameter can be inferred from those modes. For a single spatial coordinate r, the quantum Cramér–Rao bound gives

$$\mathrm{Var}(\hat{r}) \geq \frac{1}{\nu F_Q(r)}. \tag{63}$$

For several spatial parameters, the corresponding quantum Fisher information matrix determines the covariance bounds and reveals any incompatibility among their optimal measurements.

Direct intensity imaging can lose Fisher information as two incoherent sources approach one another because their overlapping intensity distributions change only weakly with separation. A measurement resolved in an appropriate spatial-mode basis can retain information that direct detection discards, even when the source light itself is classical. The resulting improvement comes from extracting the available spatial information more efficiently and should therefore be distinguished from an advantage produced by a nonclassical probe state [56].

Squeezed, entangled, or number-correlated spatial modes can provide further gains in active imaging, beam displacement estimation, and optical phase reconstruction. Their metrological value depends on propagation loss, detection efficiency, photon number, state-preparation overhead, and the measurement used at the receiver. Quantum-enhanced resolution can therefore arise from probe-state engineering, measurement design, or a combination of both, with each experiment compared against the corresponding classical and quantum estimation bounds.

Practical rule of thumb

A superresolution claim should identify the estimated parameter, optical state, measurement strategy, photon budget, and reference bound. Comparison with the Rayleigh criterion concerns resolution under a specified imaging procedure, whereas comparison with the quantum Cramér–Rao bound concerns the estimator precision permitted by the state, channel, and available measurement.

7.5 A hierarchy of experimental claims

Table 4: Evidence required for increasingly strong SQL statements.

Claim	Required comparison	Common missing element
Sub shot noise readout	Detected readout variance against a coherent or vacuum reference at equal signal resource	Loss corrected resource accounting
Imprecision below SQL scale	Calibrated input referred imprecision against a stated SQL displacement scale	Backaction and thermal noise
Quantum noise below SQL	Calibrated sum of quantum imprecision, backaction, and correlations against the same SQL convention	Classical noise and model uncertainty
Total sensitivity below SQL	Measured total estimator noise, including thermal and technical contributions, against a calibrated SQL curve	Independent calibration and uncertainty propagation
Metrological gain below separable bound	Estimator variance or Fisher information against equal resource separable probes	State preparation cost, contrast, and bias

Table 5: How the SQL enters representative platforms.

Platform	Measured parameter	Standard balance or resource restriction	References
Laser interferometer	Differential displacement or strain	Phase imprecision against radiation pressure backaction	[9, 24, 48]
Cavity optomechanics	Displacement or force	Optical imprecision against photon number backaction	[14, 19]
Atomic force microscope	Cantilever motion and force	Optical readout, cantilever disturbance, and thermal force	[50, 52]
Spin ensemble	Phase, frequency, or field	Independent projection noise against collective correlations	[54, 55]
Microwave amplifier	Field quadratures	Phase preserving gain against commutator preserving added noise	[10, 44]

Platform	Measured parameter	Standard balance or resource restriction	References
Quantum phase estimation	Optical or atomic phase	Separable generator variance against correlated generator variance	[28, 33]
Quantum imaging	Source position, separation, or another spatial parameter	Direct imaging compared with quantum-optimal spatial-mode measurements; nonclassical-probe bounds require a separately specified active-imaging model	[16, 56]

Practical rule of thumb

Reserve the phrase total sensitivity below the SQL for a comparison that includes intrinsic mechanical motion, technical noise, measurement imprecision, backaction, correlations, and calibration uncertainties in the same units and bandwidth.

8 Experimental design and numerical verification

Before beginning the numerical analysis, specify the parameter to be estimated, the estimator bandwidth, and the spectral convention. These choices define the physical problem and provide the basis for a meaningful optimization.

8.1 Noise budget and calibration sequence

A defensible SQL comparison can be built with the following sequence.

1. Define the input variable and estimator. State whether the result refers to displacement, force, strain, phase, frequency, or field.
2. Measure or model the complex transfer function from the input variable to the recorded channel. Preserve its phase.
3. Fix the Fourier normalization, spectral sidedness, symmetrization, resolution bandwidth, windowing, and averaging method.
4. Calibrate imprecision with the physical coupling suppressed or independently varied.
5. Calibrate backaction by varying probe strength and measuring the driven response or induced heating.
6. Estimate the cross spectrum with phase information. A magnitude squared coherence by itself discards the sign needed for cancellation.
7. Measure thermal and technical backgrounds and propagate their calibration uncertainty.
8. Construct the SQL from the measured effective susceptibility and the same convention used for data.

9. Repeat the comparison across probe strengths and readout quadratures to show the expected covariance geometry.

For a total measured displacement spectrum,

$$\bar{S}_{xx}^{\mathrm{meas}} = \bar{S}_{xx}^{\mathrm{intrinsic}} + \bar{S}_{xx}^{\mathrm{tech}} + \bar{S}_{xx}^{\mathrm{add}}. \tag{64}$$

The intrinsic spectrum can be decomposed into thermal and zero-point contributions when the adopted noise model treats them separately. Subtraction of a modeled background can isolate an inferred quantum contribution, as in low-frequency interferometer analyses, and the resulting claim should identify the subtraction procedure, model assumptions, and associated uncertainty.

8.2 Control parameters for an SQL heat map

The analytic heat map in Fig. 4 has clear controls. The damping ratio γ sets resonance width, the plotted frequency interval sets the mechanical response range, the range of $\log_{10}\lambda$ sets the available measurement strength, and the sample count sets rendering density. These values appear as control knobs in the preamble. A laboratory version can replace Eq. (23) with measured complex susceptibility and replace Eq. (24) with the measured covariance matrix.

Define a dimensionless excess added noise metric

$$H(\Omega,\lambda) = \log_{10}\left[\frac{\bar{S}_{xx}^{\mathrm{add}}(\Omega,\lambda)}{\bar{S}_{xx}^{\mathrm{SQL}}(\Omega)}\right]. \tag{65}$$

$H=0$ lies on the selected SQL, $H>0$ lies above it, and $H<0$ lies below it. A plot with correlations must evaluate the complex cross term in Eq. (7). Using $|\bar{S}_{xF}|$ without its phase can produce a physically incorrect valley.

8.3 Common diagnostic failures

Several checks catch common errors before a sub SQL conclusion is drawn.

- Dimensional consistency must hold after input referral. A force spectrum multiplied by $|\chi_m|^2$ has displacement spectrum units.
- The optimum of Eq. (15) must occur where its two positive terms are equal.
- Conversion from a double-sided to a one sided power spectrum multiplies by two, and the corresponding amplitude spectrum multiplies by $\sqrt{2}$.
- A squeezed variance quoted in decibels uses $10\log_{10}(V/V_0)$. An amplitude ratio uses $20\log_{10}(A/A_0)$.
- A sub SQL quantum component can coexist with total detector noise above the SQL.
- A narrowband noise minimum can differ from the setting that maximizes broadband waveform Fisher information.

Practical rule of thumb

Treat susceptibility phase, cross spectral phase, sidedness, and calibration covariance as primary data. These quantities decide whether a plotted sub SQL valley represents physical sensitivity or a convention mismatch.

9 Synthesis

The standard quantum limit acquires physical content after the measured quantity, resource allocation, probe state, estimator, bandwidth, and noise convention have been specified. An uncorrelated continuous position meter gives $\hbar|\chi_m|$ in the symmetrized double-sided convention, independent probes give the familiar inverse square root statistical scaling, and a phase preserving amplifier approaches one half quantum of added input referred noise at large gain. Correlation aware force sensing is governed by a lower dissipative constraint involving the imaginary part of the inverse susceptibility. These expressions arise from related quantum principles, although they describe distinct measurement tasks and cannot be compared without a common experimental definition.

Quantum-enhanced sensing improves precision by modifying one or more assumptions within the reference measurement model. Squeezing reshapes the probe covariance, and variational readout preserves correlations that remove part of the backaction from the estimator. Backaction evasion confines the disturbance to a conjugate mechanical quadrature, speed meters change the observable being sensed, and entanglement increases the variance of the parameter generator beyond the separable-probe value. A QND interaction preserves the observable required by later measurements. Each method retains the full uncertainty structure of the quantum system while surpassing a benchmark derived under a more restricted set of resources or couplings.

A convincing experimental result reports the benchmark, estimator, resource count, spectral convention, complete noise budget, and propagated calibration uncertainty together. This level of documentation permits meaningful comparisons among gravitational wave interferometers, optomechanical force sensors, atomic and spin ensembles, force microscopes, quantum imagers, and microwave readout systems.

Practical rule of thumb

The SQL is a calibrated design benchmark. A complete sub SQL result identifies the assumption that has changed and demonstrates the improvement using the full estimator noise budget.

A Fourier and spectral conversion reference

With

$$A(\Omega) = \int_{-\infty}^{\infty} A(t)e^{i\Omega t}\,\mathrm{d}t, \qquad A(t) = \int_{-\infty}^{\infty} \frac{\mathrm{d}\Omega}{2\pi} A(\Omega)e^{-i\Omega t}, \tag{66}$$

Eq. (4) gives the double-sided symmetrized spectrum. For a real stationary process,

$$\langle A^2(t)\rangle = \int_{-\infty}^{\infty} \frac{\mathrm{d}\Omega}{2\pi} \bar{S}_{AA}(\Omega) = \int_{0}^{\infty} \frac{\mathrm{d}\Omega}{2\pi} S^{(1)}_{AA}(\Omega). \tag{67}$$

Spectra per hertz use $f = \Omega/(2\pi)$ and satisfy $S^{[f]}(f)\,\mathrm{d}f = S^{[\Omega]}(\Omega)\,\mathrm{d}\Omega$. Stating the integration measure prevents a hidden factor of 2π.

B Analytic heat map definition

The heat map uses $r = \Omega/\Omega_m$, $\ell = \log_{10}\lambda$, and damping ratio γ. Its plotted function is

$$H(r,\ell) = \log_{10}\left[\frac{10^{-\ell} + 10^{\ell}|\widetilde{\chi}(r)|^2}{2|\widetilde{\chi}(r)|}\right]. \tag{68}$$

The arithmetic geometric mean inequality gives $H \geq 0$. Equality occurs at

$$\ell_{\mathrm{opt}}(r) = -\log_{10} |\widetilde{\chi}(r)|. \tag{69}$$

This analytic construction replaces the external `sql_heatmap.dat` dependency and compiles from the supplied TeX source and bibliography database.

C Compact formula sheet

Table 6: Frequently used formulas and their scope.

Formula	Scope
$\bar{S}_{xx}^{\mathrm{add}} = \bar{S}_{xx}^{\mathrm{imp}} + \lvert\chi_m\rvert^2 \bar{S}_{FF}^{\mathrm{ba}} + 2\,\mathrm{Re}(\chi_m^* \bar{S}_{xF})$	Linear input referred displacement measurement
$\bar{S}_{xx}^{\mathrm{SQL}} = \hbar \lvert\chi_m\rvert$	Ideal uncorrelated meter, symmetrized double-sided spectrum
$\bar{S}_{FF}^{\mathrm{SQL}} = \hbar / \lvert\chi_m\rvert$	Force referred form of the same SQL
$\bar{S}_{FF}^{\mathrm{add}} \geq \hbar \lvert \mathrm{Im}\, \chi_m^{-1} \rvert$	Correlation aware stationary dissipative force bound
$F_Q = 4(\Delta G)^2$	Pure state under unitary parameter encoding
$F_Q \leq \sum_j (\Delta g_j)^2$	Separable probes with bounded local generator ranges
$F_Q = N^2$	Ideal N qubit GHZ probe under the stated local generator
$n_{\mathrm{add}} \geq (1 - G^{-1})/2$	Deterministic phase preserving bosonic amplifier
$\xi_R^2 = N(\Delta J_\perp)^2 / \lvert\langle \mathbf{J} \rangle\rvert^2$	Metrological spin squeezing parameter

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